How-To Library

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Handling Errors That Occur When Writing Data to Remote PowerExchange Targets

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Abstract

This article describes error handling options for PowerCenter sessions that use a PowerExchange batch or relational connection to write data across a network to PowerExchange relational or nonrelational targets. It also describes the types of errors that might occur.

Supported Versions

- PowerExchange 10.1.1

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Overview

PowerCenter bulk data movement and CDC sessions can use PowerExchange (PWX) batch and relational connections to write data to PowerExchange relational and nonrelational targets over the network. PowerCenter provides several options for configuring how to handle the errors that PowerExchange might report when writing data to a target.

PowerExchange can write data to the following target types:

- DB2 for i5/OS
- DB2 for z/OS
- Nonrelational targets on z/OS, including:
  - Adabas
  - IMS synchronous
  - IMS unload files
  - Sequential data sets and tape data sets
  - VSAM ESDS, KSDS, and RRDS data sets
- Relational databases on Linux, UNIX, and Windows, such as Oracle, Microsoft SQL Server, and DB2

  Note: Informatica recommends that you use the native drivers that PowerCenter supplies to write data directly to these databases instead of using PWX connections.

In PowerCenter, you can configure the following key session properties and connection attribute to control error handling:

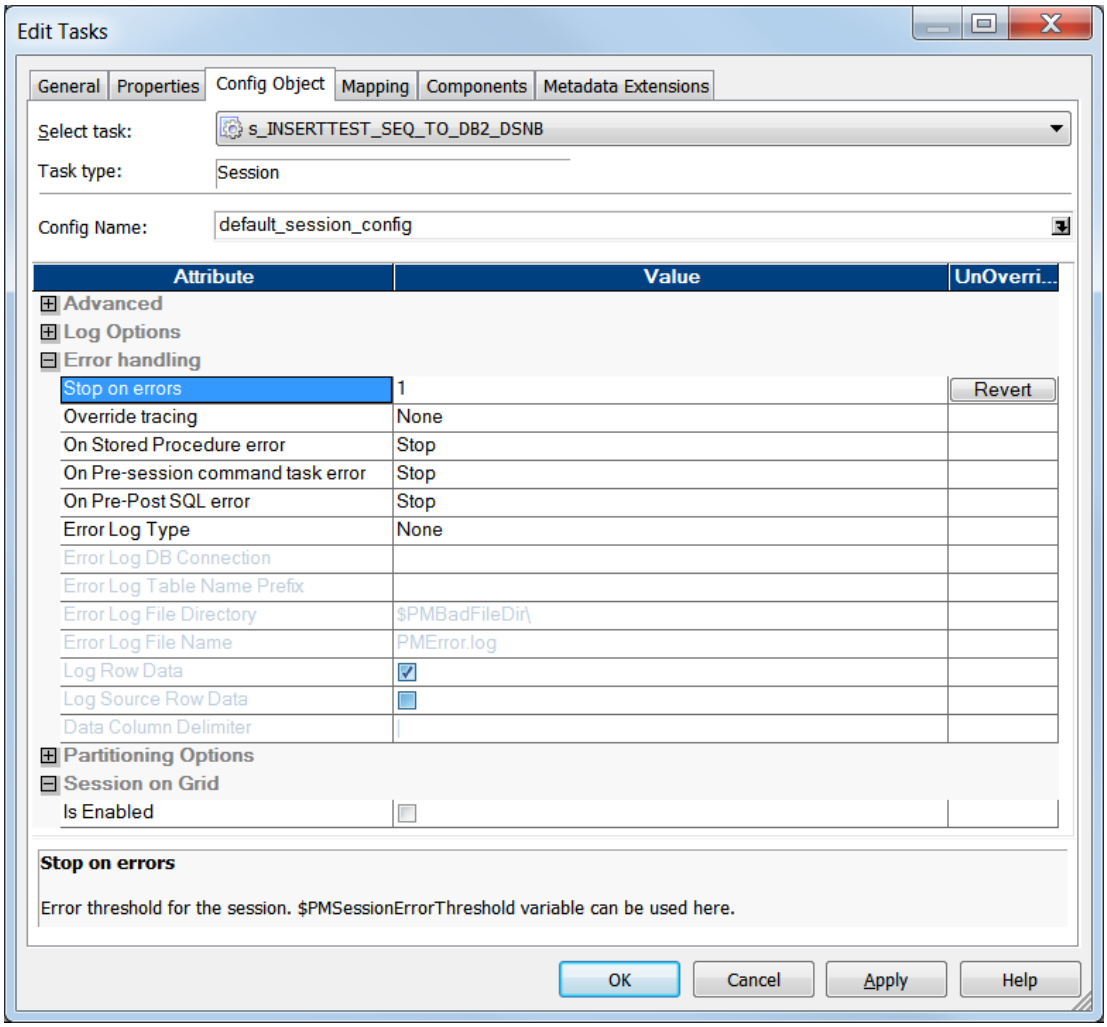
- **Stop on errors** session property
- **Error Log Type** session property

- **Write Mode** connection attribute, for a PWX batch or relational connection to a target

This article describes each of these options and the types of errors that might occur.

Stop on Errors Session Property

This session property is critical for controlling target processing and error handling. Enter the number of data records with non-fatal errors that the Integration Service can encounter before it stops the session. Non-fatal errors include reader, writer, and DTM errors. Set this property on the **Config Object** tab of the **Edit Tasks** dialog box:



Use the following criteria to set the property:

- If the session writes data to relational targets on Linux, UNIX, or Windows or to PowerExchange nonrelational targets on z/OS, enter 0. With this setting, non-fatal errors do not cause the session to stop. The session continues processing data from all sources, even if it encounters a record with an error for one of them. To determine if non-fatal errors occurred, periodically review the contents of the reject file or error log on the target.

  Note: You must stop the session to view the records with errors in the reject file. To read the reject file, you must create a custom data map. Alternatively, you can write the records to an error log database or flat file by setting the **Error Log Type** session property.

- If the session writes data to PowerExchange DB2 for i5/OS or DB2 for z/OS targets, enter a value greater than 0 to cause the workflow to abort when the specified number of errors is met. Without this error threshold, the workflow continues running and might log an excessive number of errors.
  - Enter 1 if you expect the data to be clean and the target database to *not* get duplicate-key or referential-integrity errors.
  - Otherwise, enter a large number, such as 10000, depending on the data volume that the workflow processes and the number rows that can be reprocessed.

The default value is set in the configuration that you used to create the session. The default_session_config sets **Stop on errors** to 0.

Error Log Type Session Property

You can set the **Error Log Type** session property to log any data records with errors that the writer rejected to a relational database or flat file. In PowerCenter, set this property on the **Config Object** tab of the **Edit Tasks** dialog box.

Enter one of the following options:

- **Relational Database**. Create a database table for the rejected records. If you select this option, you must also set the **Error Log DB Connection** property to specify the database connection.
- **Flat File**. Create a flat file for the rejected records. If you select this option, you can also set the **Error Log File Name** property to specify a file name other than the default name of PMError.log.
- **None**. Do not log rejected records to a relational database or flat file.

Default is **None**.

Note: If you use the **None** option and set the **Write Mode** connection attribute to **Asynchronous with Fault Tolerance**, the session writes rejected records to the reject file by default. You cannot view the reject file until the session stops. Also, the reject file is difficult to view without creating a custom data map. However, if you set the **Error Log Type** session property to write rejected records to an error log database or file, you can view the records to diagnose the errors while the session continues to run.

To flush every rejected record to the error log database or file immediately, you must also set the following custom properties for the Integration Service:

- ErrorLogDeadLockRetryOn=Yes
- ErrorLogRowBlockSize=1

For more information about error logging, see the *PowerCenter Advanced Workflow Guide*.

Write Mode Connection Attribute

When you define a PWX relational connection or NRDB Batch application connection for a target in a PowerCenter session, you can set the **Write Mode** attribute to indicate the mode in which to send data to the PowerExchange Listener. The write mode can affect error handling on the target.

Options are:

Confirm Write On

Sends data to the PowerExchange Listener and waits for a success or no success response before sending more data. This mode sends data synchronously to the PowerExchange Listener rather than buffering the data. After sending an SQL request, the sender waits for the response from PowerExchange before sending the next SQL request.

This mode provides good error recovery but has slower data transfer rates than the other modes.

Tip: To stop session execution when the session encounters non-fatal errors, specify a value greater than 0 in the **Stop on errors** session attribute on the **Config Object** tab of the **Edit Tasks** dialog box.

If you run CDC workflows with the **Resume from last checkpoint** recovery strategy and write data to targets with a PWX batch or relational connection, specify **Confirm Write On** as the write mode for the target connection.

Confirm Write Off

Sends data asynchronously to the PowerExchange Listener by buffering the data. This mode does not wait for a success or no success response. While this mode provides a faster data transfer speed than **Confirm Write On**, it does not provide the ability to determine which SQL statement failed in error situations. If an error occurs, you must reload the entire table to ensure data integrity. Use this setting only if you can reload the tables.

Caution: When you use **Confirm Write Off**, PowerCenter statistics are unreliable.

Asynchronous with Fault Tolerance

Combines the speed of **Confirm Write Off** with the error detection of **Confirm Write On**. This mode buffers the data and sends it asynchronously to the PowerExchange Listener. When an SQL error occurs, PowerExchange creates a reject file on the target machine, which contains the data records that the writer could not write to the target. View the file contents to identify and correct the errors without reloading the entire table. You can also specify how to handle specific SQL return codes.

Tip: To stop session execution when the session encounters non-fatal errors, specify a value greater than 0 in the **Stop on errors** session attribute on the **Config Object** tab of the **Edit Tasks** dialog box.

The **Asynchronous with Fault Tolerance** option is available only for PWX DB2zOS, PWX DB2i5OS, PWX DB2LUW, PWX MSSQLServer, and PWX Oracle relational connections. However, for the relational targets on Linux, UNIX, and Windows, sessions usually load data directly to the targets by using the native drivers instead of a PWX relational connection.

Default is **Confirm Write On**.

Types of Errors

PowerExchange can report the following types of errors when writing data to targets:

- Row errors that the PowerCenter Client detects before the records are sent to the target over the network
- Row errors that the target database or nonrelational interface detect
- Permanent errors, such as network, connection, and target database availability errors
- Unrecognized types of errors

Row Errors Detected by the PowerCenter Client

A PowerCenter Client a Linux, UNIX, or Windows system can detect the following types of row errors:

- During an INSERT operation, if a numeric data conversion error occurs because the incoming number has more digits than can be stored in the target column, PowerExchange issues error message PWX-00141.
- During an UPDATE operation, if a numeric data conversion error occurs because the incoming number has more digits than can be stored in the target column, PowerExchange issues error message PWX-00147.
- During an INSERT or UPDATE operation, if an incoming null data instance is being used with a column that cannot accept null data, PowerExchange issues error message PWX-00143.

The rows with errors that the PowerCenter Client detects are counted as soon as the errors occur, before the rows are sent over the network to the target.

At the time a PowerCenter Client detects a row error, a large number of valid rows might be waiting for transmission across the network to a PowerExchange Listener on z/OS or i5/OS. Depending on the maximum buffer size that is set in the APPBUFSIZE statement in the DBMOVER configuration file and the amount of data being processed, several thousand rows might be waiting.

How the system responds to these types of errors depends on the **Write Mode** connection attribute. If you use the default value of **Confirm Write On**, the session responds as follows:

1. Flushes the valid rows across the network so that they are written to the target.
2. Tells the caller which row is in error so that the error gets reported in the PowerCenter session log and reject file.
3. Either continues processing data after the problematic row or ends abnormally, depending on how the **Stop on errors** session property is configured.

If the PowerCenter Client reports illegal null row errors and you specified the CNV_ERROR_OK parameter in the **PWX Override** connection attribute, the CNV_ERROR_OK setting determines how the system responds, as follows:

- If CNV_ERROR_OK is set to Y, PowerExchange issues error message PWX-00143 and then continues writing data to the target.
- If CNV_ERROR_OK is set to N (the default value), PowerExchange treats the illegal null error that PWX-00143 reports as a permanent error and aborts the workflow.

Row Errors Detected by the Target Database or Nonrelational Interface

The target database management system (DBMS) or nonrelational interface can typically detect the following types of errors:

- Duplicate key errors that occur when an attempt is made to store a row in a table with a key value that matches an existing row but the index does not allow duplicate key values. Duplicate key errors can be reported for the following target types:
  - Adabas
  - DB2 for i5/OS
  - DB2 for z/OS
  - IMS
  - VSAM KSDS
- Referential integrity errors that occur when a parent row or record is missing

The rows with errors that the target database or nonrelational interface detects are counted as soon as they occur if the **Write Mode** connection attribute is set to **Confirm Write On**. If the **Write Mode** is set to **Confirm Write Off** or **Asynchronous with Fault Tolerance**, the count of row errors on the target machine is available later, for example, when the session ends or a commit is issued.

How the system responds to these errors depends on whether multiple rows are inserted in a single DBMS call, whether the target access method allows processing to continue after the error, and the **Write Mode** connection attribute.

The following examples demonstrate how the system responds to certain types of row errors:

Example 1. When the session writes an array of rows to a DB2 for z/OS target and DB2 returns SQL codes 252, 253, or 254

The PowerCenter session responds as follows:

1. Processes INSERT operations one at a time until the row with the error is encountered.
2. Returns control back to the caller with an indication of which row is in error. The error is then reported to the PowerCenter session log and reject file.

3. Depending on the **Stop on errors** session property setting, the session either continues processing data or ends abnormally.

Example 2. When the session writes a row to a nonrelational target on z/OS and a row error is detected

The PowerCenter session responds as follows:

1. Returns control back to the caller with an indication of which row is in error. The error is then reported in the PowerCenter session log and reject file.
2. Depending on how the **Stop on errors** session property setting, the session either continues processing data or ends abnormally.

Permanent Errors

Permanent errors include the following types of errors:

- The PowerCenter session encounters an error when trying to connect to the target database.
- The network stops working.
- A relational target database is taken offline.
- A nonrelational target file reaches its maximum extent on disk. This error is referred to as a B37 ABEND.

If the error occurs when connecting to the target database or during the prepare and describe steps, the session is aborted.

If the error occurs during an INSERT or UPDATE operation, one or more of the following results might occur:

- If the error is detected in a PowerExchange Listener subtask on z/OS or i5/OS, PowerExchange logs an error message in the range PWX-07401 to PWX-07431.
- When a fatal error is detected for the first time on Linux, UNIX, or Windows, PowerExchange logs an error message in the range PWX-07501 to PWX-07530.
- PowerExchange informs PowerCenter that a fatal error has occurred.
- The PowerCenter session is aborted if it uses an PWX application connection to write to a nonrelational target on z/OS, or it continues processing data if it uses a PWX relational connection and the error threshold that is set in the **Stop on errors** session property has not yet been exceeded .
- If the PowerCenter session continues processing data after a permanent error, PowerExchange immediately rejects the rows with errors. The rejected rows are not sent across the network to the target system. In extreme cases, this behavior might cause a large number of rejected rows to be listed in the PowerCenter session log and in the reject file.

Unrecognized Errors

If an error cannot be classified among the tolerated pre-write or DBMS row errors, PowerExchange treats the error as a permanent error and rejects all rows after the row with the error and signals that the workflow to abort.

The frequency of unrecognized errors is low. It is possible to expand the number of errors handled as pre-write or DBMS row errors.

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